

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Mechanical) (Credit System) Degree Examinations

May - June 2023

20MEE52 - WELDING TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe SMAW welding with neat sketch.	10	L*1	1	1
	b) Briefly explain Plasma arc welding process with neat sketch.	10	L2	2	1
2.	a) Explain the Tungsten Inert Gas Welding process.	10	L2	1	1
	b) Discuss the phenomenon of cold cracking and lamellar tearing in carbon steel welding.	10	L2	2	1
3.	a) Describe MIG welding with neat sketch.	10	L2	1	1
	b) Discuss laser beam welding with a neat sketch.	10	L2	2	1
Unit – II					
4.	a) Explain tensile testing on welded parts with neat sketch.	10	L1	3	1
	b) Briefly explain factors influencing fatigue life of welded joints.	10	L2	4	1
5.	a) Describe liquid penetrant testing with neat sketch.	10	L2	3	1
	b) Explain basic welding joints and symbols with neat sketch.	10	L2	4	1
6.	a) Describe Ultrasonic testing with neat sketch. Mention its advantages.	10	L2	3	1
	b) Briefly explain knowledge based expert system in welding.	10	L2	4	1
Unit – III					
7.	a) Explain safety procedures to be followed for welding operations and discuss personnel protective equipment used during welding.	10	L2	5	2
	b) Describe applications of welding in Automobile manufacturing Industries.	10	L2	5	2
8.	a) Explain welding application in ship building.	10	L2	5	1
	b) Discuss advantages and limitations of automatic welding in Industries.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
